21 Unpacking the type: attribute-level network modelling of the Lunigiana statue-stelae in R

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The Lunigiana statue-stelae (north-western Italy) are among Europe's densest and most contextually impoverished assemblages of prehistoric anthropomorphic sculpture. Our previous bipartite site–type model showed Ambrosi's (1972) typological classes holding stable, non-random structural positions. Ties, however, were derived deterministically from shared class, so the QAP correlation was tautological by construction (r = 1.000); and re-auditing the deposited dataset showed its 61 site nodes resolving to 49 places, with the network's entire bridging structure running through the residual class of unattributable stelae. The flaw surfaced only because the data were open.

We rebuild the corpus — controlled site vocabulary with declared spatial granularity, coordinates recomputed from the original UTM data — and decompose the typology across 76 Chalcolithic monuments, setting aside five reworked Iron Age Type C stelae. The residual class is itself a result: of the eleven specimens the typology failed to classify, one has no object behind it, three were already in the dataset under another class, and the rest are now identified. Every specimen is coded against some forty morphological and iconographic attributes, under a three-state scheme separating non-observable from absent ones. Typological class is held out of the similarity computation, used only as a post hoc label.

The workflow is entirely FLOS and reproducible in R: Gower similarity on mixed data (cluster::daisy), backbone extraction (backbone), stochastic blockmodelling against Louvain (sbm), ERGM and QAP with substantive nodal covariates (statnet, sna), and multiple imputation of unobservable attributes, with interval reporting. A sensitivity test reruns every metric at both site scales.

The question is whether a legacy typology encodes relational structure or imposes it. Dataset, normalisation log, pre-registered hypotheses and code are released openly, with an explicit diff against the earlier deposit